Biological Methods for Engineers (227-0949-00L) - Mock Exam Protocols, Autumn Semester 2025

Combined question sheet and scoring rubric generated from JSON source. Criterion headings match rubric_manifest.json.

q1 - Section 1: Conceptual & Mechanistic Questions - Cell Culture Question... - 1 point:

Question (q1): Section 1: Conceptual & Mechanistic Questions - Cell Culture

Question 1(a): Explain why CHO cells are commonly used in biotechnology. Refer to at least two biological characteristics that make them advantageous.
Reference answer: CHO cells are robust, easy to grow, and capable of producing very high levels of recombinant proteins. Their mammalian origin allows them to perform post-translational modifications and proper folding of secreted proteins, making them the workhorse of industrial biomanufacturing.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q2 - Section 1: Conceptual & Mechanistic Questions - Cell Culture Question... - 1 point:

Question (q2): Section 1: Conceptual & Mechanistic Questions - Cell Culture

Question 1(b): Describe what happens to adherent mammalian cells during trypsinization at the molecular level.
Reference answer: Trypsin cleaves extracellular matrix proteins and adhesion molecules such as integrins that normally anchor cells to the dish. As these proteins are digested, the cells round up and detach. Prolonged exposure can damage membrane proteins, but under standard conditions cells remain viable.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q3 - Section 1: Conceptual & Mechanistic Questions - Cell Culture Question... - 1 point:

Question (q3): Section 1: Conceptual & Mechanistic Questions - Cell Culture

Question 1(c): During splitting, the medium must be warmed to 37 °C. Explain two consequences of using medium that is too cold or overheated.
Reference answer: Medium that is too cold shocks mammalian cells, inhibiting adhesion, slowing metabolic activity, and delaying recovery after splitting. Overheated medium degrades heat-sensitive growth factors and can harm the cells. Medium warmed to 37 °C avoids both issues.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q4 - Section 1: Conceptual & Mechanistic Questions - Cell Culture Question... - 1 point:

Question (q4): Section 1: Conceptual & Mechanistic Questions - Cell Culture

Question 1(d): Define "contact inhibition." Why does subculturing prevent senescence?
Reference answer: Contact inhibition is the cessation of cell division upon reaching a high density and making sustained contact with neighboring cells. Subculturing prevents cells from entering senescence by maintaining them below confluency and allowing continued proliferation.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q5 - Section 1: Conceptual & Mechanistic Questions - Protein Isolation & B... - 1 point:

Question (q5): Section 1: Conceptual & Mechanistic Questions - Protein Isolation & Bradford Assay

Question 2(a): In the Bradford assay, why does Coomassie G-250 shift its absorbance when bound to protein?
Reference answer: Coomassie G-250 undergoes a spectral shift when binding proteins, stabilizing its blue anionic form, which absorbs strongly at 595 nm. This shift forms the basis of the Bradford assay.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q6 - Section 1: Conceptual & Mechanistic Questions - Protein Isolation & B... - 1 point:

Question (q6): Section 1: Conceptual & Mechanistic Questions - Protein Isolation & Bradford Assay

Question 2(b): Explain why detergents like SDS disrupt protein–protein interactions, whereas nonionic detergents often do not.

Reference answer: SDS is an ionic detergent with a charged headgroup that disrupts protein–protein interactions and fully denatures proteins. Nonionic detergents like Triton X-100 lack charge and therefore solubilize membranes while preserving most protein complexes.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q7 - Section 1: Conceptual & Mechanistic Questions - Protein Isolation & B... - 1 point:

Question (q7): Section 1: Conceptual & Mechanistic Questions - Protein Isolation & Bradford Assay

Question 2(c): A student accidentally adds 0.2% Triton X-100 to their Bradford assay sample. Predict what will happen to their absorbance readings and why.
Reference answer: Triton X-100 increases background absorbance because it interacts with the dye micelles, shifting the baseline even without protein. As a result, protein concentration will be falsely overestimated.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q8 - Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprec... - 1 point:

Question (q8): Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprecipitation

Question 3(a): Why does boiling samples in LDS/SDS sample buffer help proteins migrate according to size?
Reference answer: Boiling with SDS and reducing agents denatures proteins completely and breaks disulfide bonds. SDS coats proteins with a uniform negative charge, ensuring migration through the gel is determined solely by size.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q9 - Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprec... - 1 point:

Question (q9): Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprecipitation

Question 3(b): In immunoprecipitation of Actin using magnetic beads: describe the purpose of the wash steps, and explain the consequence of washing too harshly vs. too gently.
Reference answer: Wash steps remove unbound and nonspecifically bound proteins from the beads. Excessively harsh washing can strip the target protein off the beads, while overly gentle washing leaves contaminants that appear as background bands.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q10 - Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprec... - 1 point:

Question (q10): Section 1: Conceptual & Mechanistic Questions - SDS-PAGE & Immunoprecipitation

Question 3(c): You load 20 µg and 5 µg of egg-white lysate on a gel. Describe how the banding patterns would differ, and why.
Reference answer: A 20 µg load shows darker, thicker bands and may reveal faint low-abundance proteins. A 5 µg load yields cleaner but less sensitive banding.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q11 - Section 1: Conceptual & Mechanistic Questions - qRT-PCR Question 4(a)... - 1 point:

Question (q11): Section 1: Conceptual & Mechanistic Questions - qRT-PCR

Question 4(a): Explain the purpose of using a poly-T primer during reverse transcription.
Reference answer: Poly-T primers bind the poly-A tails of mature mRNA, ensuring selective reverse transcription of mRNA while excluding rRNA and tRNA.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q12 - Section 1: Conceptual & Mechanistic Questions - qRT-PCR Question 4(b)... - 1 point:

Question (q12): Section 1: Conceptual & Mechanistic Questions - qRT-PCR

Question 4(b): Define CT (cycle threshold). Why must it be measured during the exponential phase?
Reference answer: The cycle threshold (CT) is the PCR cycle at which fluorescence surpasses background. It must be measured during the exponential phase to ensure linearity between product quantity and fluorescence.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q13 - Section 1: Conceptual & Mechanistic Questions - qRT-PCR Question 4(c)... - 1 point:

Question (q13): Section 1: Conceptual & Mechanistic Questions - qRT-PCR

Question 4(c): A sample treated with tunicamycin shows a CT value for GRP94 that is 3 cycles lower than the DMSO control. [The original exam gives no explicit question here; interpret this result.]
Reference answer: A CT that is three cycles lower indicates significantly more GRP94 mRNA in the tunicamycin-treated sample, reflecting ER stress-induced transcriptional upregulation.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q14 - Section 1: Conceptual & Mechanistic Questions - qRT-PCR Question 4(d)... - 1 point:

Question (q14): Section 1: Conceptual & Mechanistic Questions - qRT-PCR

Question 4(d): Why must GRP94 expression be normalized to a housekeeping gene (such as PABPN1 or GAPDH)?
Reference answer: A housekeeping gene normalizes for differences in RNA input, cDNA synthesis efficiency, and pipetting variability. Its expression should remain constant across conditions.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q15 - Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis &... - 1 point:

Question (q15): Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis & Splicing

Question 5(a): Explain why a 2.5% agarose gel is used to resolve spliced vs. unspliced XBP-1 PCR products.
Reference answer: Spliced and unspliced XBP-1 PCR products differ by only 26 bp. A 2.5% agarose gel has small pores that provide the high resolution needed to separate fragments with such small size differences.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q16 - Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis &... - 1 point:

Question (q16): Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis & Splicing

Question 5(b): A group sees only a single band for XBP-1 after tunicamycin treatment. List two possible biological explanations and two technical explanations.
Reference answer: Biological: (1) incomplete or weak XBP1 splicing; (2) ineffective ER stress induction. Technical: (1) poor PCR amplification or primer issues; (2) gel problems such as incorrect percentage or allowing the smaller band to run off.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q17 - Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis &... - 1 point:

Question (q17): Section 1: Conceptual & Mechanistic Questions - DNA Electrophoresis & Splicing

Question 5(c): Describe how loading dye helps ensure successful gel electrophoresis.
Reference answer: Loading dye adds density, allowing the sample to sink into the well, and provides colored tracking dyes that move through the gel to monitor electrophoresis progress. It does not stain DNA.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q18 - Section 2: Data Interpretation - qPCR Data Interpretation You obtain... - 1 point:

Question (q18): Section 2: Data Interpretation - qPCR Data Interpretation
You obtain the following CT values:
Condition | GRP94 CT | PABPN1 CT
DMSO | 25.4 | 20.0
Tunicamycin | 22.0 | 20.1

Question 6(a): Explain conceptually (not numerically) how the ΔΔCT method would use these values to determine fold-change.
Reference answer: The ΔΔCT method first normalizes GRP94 CT to PABPN1 CT in each condition (yielding ΔCT). The difference between treated and control ΔCT values (ΔΔCT) reflects relative expression. A lower ΔΔCT corresponds to higher fold-change.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q19 - Section 2: Data Interpretation - qPCR Data Interpretation You obtain... - 1 point:

Question (q19): Section 2: Data Interpretation - qPCR Data Interpretation
You obtain the following CT values:
Condition | GRP94 CT | PABPN1 CT
DMSO | 25.4 | 20.0
Tunicamycin | 22.0 | 20.1

Question 6(b): How would you know whether PABPN1 was a valid housekeeping gene in this dataset?
Reference answer: A valid housekeeping gene shows nearly identical CT values across conditions. Here, PABPN1 has CT values of 20.0 and 20.1, indicating stable expression.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q20 - Section 2: Data Interpretation - Electrophoresis Interpretation You r... - 1 point:

Question (q20): Section 2: Data Interpretation - Electrophoresis Interpretation
You run PCR products for XBP-1 on a 2.5% gel and obtain: two bands for tunicamycin-treated cells (strong lower band, faint upper band); one strong upper band for DMSO.

Question 7(a): Explain what each band corresponds to.
Reference answer: The upper band represents unspliced XBP1; the lower band represents spliced XBP1 (26 bp shorter). Tunicamycin induces ER stress, increasing splicing and yielding a strong lower band.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q21 - Section 2: Data Interpretation - Electrophoresis Interpretation You r... - 1 point:

Question (q21): Section 2: Data Interpretation - Electrophoresis Interpretation
You run PCR products for XBP-1 on a 2.5% gel and obtain: two bands for tunicamycin-treated cells (strong lower band, faint upper band); one strong upper band for DMSO.

Question 7(b): What could a faint lower band in DMSO-treated cells indicate about basal ER stress?
Reference answer: A faint lower band in DMSO indicates basal ER stress, as low-level IRE1 activity produces occasional XBP1 splicing even in unstressed cells.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q22 - Section 2: Data Interpretation - SDS-PAGE & IP Interpretation On a si... - 1 point:

Question (q22): Section 2: Data Interpretation - SDS-PAGE & IP Interpretation
On a silver-stained IP gel, the eluate lane shows a strong band at ~42 kDa and several faint additional bands.

Question 8(a): Explain why faint extra bands may appear even if the IP was successful.
Reference answer: Faint extra bands arise from nonspecific binders, antibody fragments, or proteins weakly interacting with beads or antibody. IP eluates are rarely perfectly pure.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q23 - Section 2: Data Interpretation - SDS-PAGE & IP Interpretation On a si... - 1 point:

Question (q23): Section 2: Data Interpretation - SDS-PAGE & IP Interpretation
On a silver-stained IP gel, the eluate lane shows a strong band at ~42 kDa and several faint additional bands.

Question 8(b): Suggest two troubleshooting steps to reduce these nonspecific bands.
Reference answer: Reduce antibody or bead amounts, increase wash stringency, or pre-clear the lysate to decrease nonspecific background.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q24 - Section 3: Practical / Troubleshooting - Cell Culture Troubleshooting... - 1 point:

Question (q24): Section 3: Practical / Troubleshooting - Cell Culture Troubleshooting

Question 9: Your CHO cells look rounded and refuse to attach after splitting. Give two possible procedural mistakes and how to correct each.
Reference answer: (1) Residual trypsin continuing to digest adhesion proteins - corrected by thorough neutralization and washing; (2) using cold medium, which shocks cells - corrected by prewarming medium to 37 °C.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q25 - Section 3: Practical / Troubleshooting - qPCR Troubleshooting During... - 1 point:

Question (q25): Section 3: Practical / Troubleshooting - qPCR Troubleshooting
During qPCR, your melting curve shows two distinct peaks.

Question 10(a): Give two reasons this can occur.
Reference answer: Two melting peaks indicate (1) nonspecific PCR products or (2) primer dimers.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q26 - Section 3: Practical / Troubleshooting - qPCR Troubleshooting During... - 1 point:

Question (q26): Section 3: Practical / Troubleshooting - qPCR Troubleshooting
During qPCR, your melting curve shows two distinct peaks.

Question 10(b): Describe one step you could take to confirm primer specificity.
Reference answer: Running the qPCR product on an agarose gel verifies whether a single specific band is present.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q27 - Section 3: Practical / Troubleshooting - IP Troubleshooting Your Acti... - 1 point:

Question (q27): Section 3: Practical / Troubleshooting - IP Troubleshooting
Your Actin IP shows no band in the eluate.

Question 11(a): List three potential causes spanning: antibody/bead issues; lysate issues; wash/elution issues.
Reference answer: Antibody/bead issues: poor antibody quality or failure to bind beads. Lysate issues: wrong detergent or insufficient protein concentration. Wash/elution issues: overly harsh washing removing the target protein.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q28 - Section 3: Practical / Troubleshooting - Induced Neurons Question 12(... - 1 point:

Question (q28): Section 3: Practical / Troubleshooting - Induced Neurons

Question 12(a): Why must iNeurons be thawed quickly but gently?
Reference answer: iNeurons must be thawed quickly to minimize ice crystal formation but handled gently because they are fragile immediately after thawing.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.

q29 - Section 3: Practical / Troubleshooting - Induced Neurons Question 12(... - 1 point:

Question (q29): Section 3: Practical / Troubleshooting - Induced Neurons

Question 12(b): During live/dead staining, what happens at the molecular level that makes Calcein AM fluoresce green in live cells and Ethidium homodimer fluoresce red in dead cells?
Reference answer: Calcein-AM enters live cells and is cleaved by esterases into fluorescent calcein (green). Ethidium homodimer-1 enters only dead cells (it cannot cross intact membranes), binds DNA, and fluoresces red.
Award up to the maximum for demonstrated correct content; use partial credit when appropriate.